Actuarial Exam P Master Formula Chart

Day 1: Sample Spaces, Set Operations, & Event Algebra

1. Fundamental Set Operations

- **Union ($A \cup B$):** Event A occurs, OR event B occurs, OR both occur.
  - *Verbal Description:* The collection of all outcomes that belong to event A, event B, or both.
- **Intersection ($A \cap B$):** Event A AND event B occur simultaneously.
  - *Verbal Description:* The collection of outcomes shared in common by both event A and event B.
- **Complement (A^c or A'):** Event A does NOT occur.
  - *Verbal Description:* All outcomes in the sample space that lie outside of event A.
- **Set Difference ($A \setminus B = A \cap B^c$):** Event A occurs, BUT event B does NOT occur.
  - *Verbal Description:* The collection of outcomes that belong exclusively to event A and not to event B.
- **Null / Empty Event ($\varnothing$):** An impossible event containing no outcomes.
  - *Verbal Description:* The set with zero outcomes, representing an event that cannot occur.

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2. De Morgan's Laws

- **Complement of Union:** $(A \cup B)^c = A^c \cap B^c$
  - *Verbal Description:* The event that neither A nor B occurs equals A not occurring AND B not occurring simultaneously.
- **Complement of Intersection:** $(A \cap B)^c = A^c \cup B^c$
  - *Verbal Description:* The event that A and B do not both occur equals A failing to occur OR B failing to occur.

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3. Set Distributive & Partition Rules

- **Distributive Laws:**
  - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
    - *Verbal Description:* The intersection of A with a union equals the union of A's intersections with each set.
  - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
    - *Verbal Description:* The union of A with an intersection equals the intersection of A's unions with each set.
- **Partitioning an Event:**

- $A = (A \cap B) \cup (A \cap B^c)$
  - *Verbal Description:* Event A can be decomposed into two mutually exclusive parts: A occurring with B, and A occurring without B.

Day 2: Axioms of Probability & Basic Probability Rules

1. Kolmogorov's Axioms of Probability

- **Axiom 1 (Non-negativity):** $P(A) \geq 0$
  - *Verbal Description:* Every event has a probability that is non-negative; probabilities can never be negative.
- **Axiom 2 (Normalization / Certainty):** $P(S) = 1$
  - *Verbal Description:* The total probability of the entire sample space is always equal to 1.
- **Axiom 3 (Countable Additivity):** For mutually exclusive (disjoint) events $A_1, A_2, \ldots$ where $A_i \cap A_j = \varnothing$ for $i \neq j$:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

  - *Verbal Description:* For any countable sequence of disjoint events, the probability that at least one occurs is the sum of their individual probabilities.
- **Finite Additivity:** $P(A_1 \cup A_2 \cup \cdots \cup A_n) = P(A_1) + P(A_2) + \cdots + P(A_n)$
  - *Verbal Description:* For a finite collection of mutually exclusive events, the probability of their union equals the sum of their individual probabilities.

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2. Derived Properties & Basic Rules

- **Empty Set Rule:** $P(\varnothing) = 0$
  - *Verbal Description:* An impossible event containing no outcomes has a probability of exactly zero.
- **Complement Rule:** $P(A^c) = 1 - P(A)$
  - *Verbal Description:* The probability that event A does not occur equals 1 minus the probability that A does occur.
- **Boundedness:** $0 \leq P(A) \leq 1$
  - *Verbal Description:* All event probabilities are bounded between 0 (impossible) and 1 (certain).
- **Difference Rule:** $P(B \setminus A) = P(B \cap A^c) = P(B) - P(A)$ for $A \subseteq B$
  - *Verbal Description:* If event A is entirely contained within event B, the probability of B occurring without A is $P(B)$ minus $P(A)$.
- **Monotonicity:** If $A \subseteq B$, then $P(A) \leq P(B)$

- *Verbal Description:* A subset event cannot have a higher probability than the larger event containing it.

- **Law of Total Probability for Sets (Partitioning):** $P(A) = P(A \cap B) + P(A \cap B^c)$
  - *Verbal Description:* The total probability of event A is the sum of the probability of A happening with B and A happening without B.

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3. Additive Laws & Set Difference Shortcuts

- **General Addition Rule (Two Events):**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

  - *Verbal Description:* The probability that A or B occurs is the sum of their individual probabilities minus the double-counted intersection probability.

- **Union Decomposition:**

$$P(A \cup B) = P(A) + P(B \cap A^c)$$

  - *Verbal Description:* The probability of A or B equals the probability of A plus the probability of B occurring exclusively without A.

- **Shortcut Set Difference Formulas:**
  - $P(B \cap A^c) = P(A \cup B) - P(A)$
    - *Verbal Description:* The probability of B only equals the combined union probability minus $P(A)$.
  - $P(A \cap B^c) = P(A \cup B) - P(B)$
    - *Verbal Description:* The probability of A only equals the combined union probability minus $P(B)$.

- **Symmetric Difference Formula:**

$$P(A \Delta B) = P(A) + P(B) - 2P(A \cap B)$$

  - *Verbal Description:* The probability that *exactly one* of event A or event B occurs, excluding the outcome where both occur.

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4. Intersection Bounds & Bonferroni's Inequality

- **Intersection Bounds (Two Events):**
  - **Minimum Possible Intersection:**

$$P(A \cap B) \geq P(A) + P(B) - 1$$

    (Note: $P(A \cap B) = P(A) + P(B) - P(A \cup B)$ *when* $P(A \cup B) = 1$*)*
    - *Verbal Description:* The smallest possible intersection probability occurs when the union spans the entire sample space ($P(A \cup B) = 1$).
  - **Maximum Possible Intersection:**

$$P(A \cap B) \leq \min(P(A), P(B))$$

    - *Verbal Description:* The maximum intersection probability is restricted by the smaller of the two individual event probabilities.

- **Bonferroni's Inequality:**
    - **Two-Event Case:**

$$P(A \cap B) \geq P(A) + P(B) - 1 \quad \text{or} \quad P(A \cap B) \geq 1 - P(A^c) - P(B^c)$$

        - *Verbal Description:* The simultaneous occurrence of two events is at least 1 minus the sum of their individual complements' probabilities.
    - **Three-Event Case ($n = 3$):**

$$P(A \cap B \cap C) \geq P(A) + P(B) + P(C) - 2$$

        - *Verbal Description:* Lower bound for the simultaneous occurrence of three events, obtained by subtracting 2 from the sum of their probabilities.
    - **General n-Event Form:**

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) \geq \sum_{i=1}^{n} P(A_i) - (n-1)$$

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) \geq 1 - \sum_{i=1}^{n} P(A_i^c)$$

        - *Verbal Description:* Lower bound for the simultaneous occurrence of n events, derived by subtracting $(n-1)$ from the sum of individual probabilities, or subtracting the sum of all complement probabilities from 1.

Day 3: Combinatorics, Permutations, & Combinations

1. Fundamental Counting Principles

- **Multiplication Principle (Sequential Choices):**

$$\text{Total Outcomes} = n_1 \times n_2 \times \cdots \times n_k$$

    - *Verbal Description:* If a procedure consists of k sequential steps with n_i options at step i, the total number of combined outcomes is the product of options at each step.

- **Addition Principle (Disjoint Alternatives):**

$$\text{Total Options} = n_1 + n_2 + \cdots + n_m$$

    - *Verbal Description:* If a single selection can be made from one of m mutually exclusive sets, the total number of options is the sum of sizes of each set.

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2. Permutations (Order Matters)

- **Basic Permutation Formula:**

$$P(n, r) = {}_nP_r = \frac{n!}{(n-r)!}$$

(where $n! = n \times (n-1) \times \cdots \times 2 \times 1$ and $0! = 1$)

- *Verbal Description:* The number of distinct ordered arrangements of r items chosen without replacement from a set of n distinct items.

- **Permutations with Repeated / Indistinguishable Objects:**

$$\text{Distinct Arrangements} = \frac{n!}{n_1!\, n_2!\, \dots\, n_k!}$$

(where $n_1 + n_2 + \cdots + n_k = n$)

- *Verbal Description:* The number of distinct linear arrangements of n total items when the items belong to k groups of identical/indistinguishable objects.

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3. Combinations (Order Does NOT Matter)

- **Combination Formula:**

$$C(n, r) = \binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!}$$

- *Verbal Description:* The number of ways to choose an unordered subset or committee of r items from n distinct available items.

- **Useful Algebraic Identities:**
  - **Symmetry:** $\binom{n}{r} = \binom{n}{n-r}$
    - *Verbal Description:* Selecting r items to include in a group is identical to selecting $n - r$ items to exclude.
  - **Boundary Values:** $\binom{n}{0} = \binom{n}{n} = 1$
    - *Verbal Description:* There is exactly 1 way to select zero items (empty set) or all n items.
  - **Pascal's Identity:** $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$
    - *Verbal Description:* The number of ways to choose r items from n equals combinations including a specific item plus combinations excluding that item.

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4. Classical Probability with Combinatorics

- **Equally Likely Outcomes Formula:**

$$P(E) = \frac{n(E)}{n(S)} = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$$

- *Verbal Description:* In a sample space of equally likely outcomes, the probability of an event equals favorable outcomes divided by total outcomes.

Day 4: Advanced Combinatorics Problem-Solving

1. Proximity Rules: Block vs. Gap Methods

- **Block Method (Items MUST sit together):**

$$\text{Total Ways} = (\text{Arrangements of Super-Items}) \times (\text{Internal Arrangements inside Block})$$

- *Verbal Description:* Treat the restricted items as a single "super-item" block, arrange all super-items, and multiply by internal arrangements inside the block.

- **Gap / Separator Method (NO TWO items can sit together):**

$$\text{Total Ways} = m! \times \binom{m+1}{r} \times r!$$

(where m is the number of unrestricted items and r is the number of restricted items)

- *Verbal Description:* First arrange the m unrestricted items in $m!$ ways, then choose r gaps out of $m+1$ available spaces for the restricted items, and multiply by $r!$ internal permutations.

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2. Distribution Problems ("Stars and Bars")

- **Non-Negative Integer Distribution (Bins CAN be empty, $x_i \geq 0$):**

$$\text{Ways} = \binom{n+k-1}{k-1} = \binom{n+k-1}{n}$$

- *Verbal Description:* The number of distinct ways to distribute n identical items into k distinct bins when bins are allowed to receive zero items.

- **Positive Integer Distribution (Bins CANNOT be empty, $x_i \geq 1$):**

$$\text{Ways} = \binom{n-1}{k-1}$$

- *Verbal Description:* The number of distinct ways to distribute n identical items into k distinct bins when every bin must receive at least 1 item.

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3. Constrained Committee Selection

- **Complement Method ("Cannot Both Serve"):**

$$\text{Valid Selections} = \binom{N}{r} - \binom{N-2}{r-2}$$

- *Verbal Description:* Calculate total unrestricted ways to select r people from N, then subtract prohibited committees where both conflicting individuals are placed together.

- **Must Include / Exclude Constraints:**

$$\text{Valid Selections} = \binom{N-a-b}{r-a}$$

(where a individuals must be included and b individuals must be excluded)

- *Verbal Description:* Reduce the required committee size by a (since a slots are pre-filled) and reduce the candidate pool by removing both the a included and b excluded individuals.

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4. Grid / Lattice Paths

- **Unrestricted Grid Paths from $(0,0)$ to (a,b):**

$$\text{Total Paths} = \binom{a+b}{a} = \binom{a+b}{b} = \frac{(a+b)!}{a!\,b!}$$

  - *Verbal Description:* The number of distinct paths moving only Right ($+x$) and Up ($+y$) equals choosing a Right steps (or b Up steps) out of $a+b$ total steps.

- **Paths Passing Through Waypoint (x_1, y_1) to Destination (a,b):**

$$\text{Paths via Waypoint} = \binom{x_1+y_1}{x_1} \times \binom{(a-x_1)+(b-y_1)}{a-x_1}$$

  - *Verbal Description:* Multiply the number of grid paths from origin $(0,0)$ to waypoint (x_1, y_1) by the number of grid paths from (x_1, y_1) to final destination (a,b).

Day 5: Mutually Exclusive Events & Addition Rules

1. Mutually Exclusive (Disjoint) Events

- **Intersection Condition:**

$$A \cap B = \varnothing \implies P(A \cap B) = 0$$

  - *Verbal Description:* Two events are mutually exclusive if they share no outcomes in common, meaning they cannot occur simultaneously and their joint probability is zero.

- **Special Addition Rule (Disjoint Events):**

$$P(A \cup B) = P(A) + P(B)$$

  - *Verbal Description:* If two events cannot overlap, the probability that at least one occurs is simply the sum of their individual probabilities.

- **Finite Additivity (n Disjoint Events):**

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) = \sum_{i=1}^{n} P(A_i)$$

  - *Verbal Description:* For any finite collection of mutually exclusive events, the probability that at least one occurs is the sum of all individual event probabilities.

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2. General Addition Rule (Two Events)

- **Two-Event General Addition Formula:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

  - *Verbal Description:* The probability that at least one of two overlapping events occurs is the sum of their individual probabilities minus the double-counted joint intersection probability.

- **Rearranged Form for Intersection:**

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

- *Verbal Description:* The probability of two events occurring together equals the sum of their individual probabilities minus the probability of their union.

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3. Inclusion-Exclusion Principle (Three Events)

- **Three-Event Union Formula:**

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

  - *Verbal Description:* The probability that at least one of three events occurs equals the sum of individual probabilities, minus the three pairwise intersections to fix double-counting, plus the triple intersection to restore the subtracted center region.

- **Sum of Pairwise Intersections:**

$$P(A \cap B) + P(A \cap C) + P(B \cap C) = P(A) + P(B) + P(C) - P(A \cup B \cup C) + P(A \cap B \cap C)$$

  - *Verbal Description:* Expresses the total sum of pairwise joint probabilities using individual probabilities, the overall union probability, and the triple intersection probability.

Day 6: Independent Events & Joint Probabilities

1. Formal Definition of Independent Events

- **Product Rule for Two Independent Events:**

$$P(A \cap B) = P(A) \cdot P(B)$$

  - *Verbal Description:* Two events A and B are independent if and only if the probability of their joint occurrence equals the product of their individual marginal probabilities.

- **Equivalent Conditional Probability Statements:**

$$P(A \mid B) = P(A) \quad \text{and} \quad P(B \mid A) = P(B) \quad (\text{for } P(A) > 0,\, P(B) > 0)$$

  - *Verbal Description:* The conditional probability of A given B equals the unconditional probability of A (and vice versa), meaning the occurrence of one event provides no information about the occurrence of the other.

- **Independence vs. Mutual Exclusivity Distinction:**

  - For non-trivial events ($P(A) > 0, P(B) > 0$), if A and B are mutually exclusive ($P(A \cap B) = 0$), then they **CANNOT** be independent because $P(A \cap B) \neq P(A)P(B)$.
  - *Verbal Description:* Mutually exclusive events are strongly dependent because knowing one event occurred guarantees the other did not occur.

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2. Independence of Complements

- **Complement Independence Rules:**

  - $P(A \cap B^c) = P(A) \cdot P(B^c)$

    - *Verbal Description:* If A and B are independent, A occurring and B not occurring are also independent events.
  - $P(A^c \cap B) = P(A^c) \cdot P(B)$
    - *Verbal Description:* If A and B are independent, A not occurring and B occurring are also independent events.
  - $P(A^c \cap B^c) = P(A^c) \cdot P(B^c)$
    - *Verbal Description:* If A and B are independent, the non-occurrence of A and non-occurrence of B are also independent events.

- **"At Least One" Strategy for Independent Events:**
  - **Two Events:**

$$P(A \cup B) = 1 - P(A^c \cap B^c) = 1 - P(A^c)P(B^c)$$

    - *Verbal Description:* The probability that at least one of two independent events occurs equals 1 minus the product of their complement probabilities.
  - **General n Independent Events:**

$$P(\text{At least one event occurs}) = 1 - \prod_{i=1}^{n} P(A_i^c) = 1 - \prod_{i=1}^{n} (1 - P(A_i))$$

    - *Verbal Description:* The probability that at least one of n independent events occurs equals 1 minus the product of all individual non-occurrence probabilities.

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3. Extension to $n \geq 3$ Events: Pairwise vs. Mutual Independence

- **Mutual Independence for Three Events (A, B, C):** Requires all 4 conditions to hold simultaneously:
  1. **Pairwise Independence (3 conditions):**

$$P(A \cap B) = P(A)P(B)$$
$$P(A \cap C) = P(A)P(C)$$
$$P(B \cap C) = P(B)P(C)$$

  2. **Triple Product Condition (1 condition):**

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

  - *Verbal Description:* Three events are mutually independent if and only if every pair is pairwise independent AND the joint probability of all three equals the product of all three individual probabilities.

- **Mutual Independence for n Events:** Requires $2^n - n - 1$ product equations to hold for all sub-collections of events:

$$P\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} P(A_j) \quad \text{for any subset } J \subseteq \{1, 2, \ldots, n\} \text{ with } |J| \geq 2$$

  - *Verbal Description:* n events are mutually independent if the joint probability of any subset of 2 or more events equals the product of their individual probabilities.

Day 7: Multiplication Rules & Multi-Step Probability Calculations

1. General Multiplication Rule

- **General Multiplication Rule (Two Dependent Events):**

$$P(A \cap B) = P(A) \cdot P(B \mid A) \quad \text{or} \quad P(A \cap B) = P(B) \cdot P(A \mid B)$$

  - *Verbal Description:* The probability that two sequential or dependent events both occur equals the probability of the first event multiplied by the conditional probability of the second event given that the first event has occurred.

- **Independent Events Special Case:**

$$P(A \cap B) = P(A) \cdot P(B)$$

  - *Verbal Description:* When events are independent, $P(B \mid A) = P(B)$, simplifying the conditional multiplication rule to a standard product of individual probabilities.

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2. The Chain Rule for n Sequential Events

- **General Chain Rule (n Events):**

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1) \cdot P(A_2 \mid A_1) \cdot P(A_3 \mid A_1 \cap A_2) \cdots P(A_n \mid A_1 \cap A_2 \cap \cdots \cap A_{n-1})$$

  - *Verbal Description:* The joint probability that a sequence of n dependent events all occur equals the probability of the first event, multiplied by the conditional probability of each subsequent event given that all preceding events in the sequence have occurred.

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3. Sampling With vs. Without Replacement

- **Sampling With Replacement (Independent Draws):**

$$P(A_1 \cap A_2 \cap \cdots \cap A_k) = [P(A)]^k$$

  - *Verbal Description:* Draws are independent because items are returned to the pool prior to the next draw, keeping probabilities constant across every stage.

- **Sampling Without Replacement (Dependent Draws / Sequential Chain Rule):**

$$P(A_1 \cap A_2 \cap \cdots \cap A_k) = \frac{n(A)}{N} \times \frac{n(A)-1}{N-1} \times \cdots \times \frac{n(A)-k+1}{N-k+1}$$

  - *Verbal Description:* Draws are dependent because items are removed without replacement, decreasing both the target item count and total pool size by 1 after each successful draw.

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4. Multi-Step Probability Pathways (Law of Total Probability Decomposition)

- **Pathway Addition Rule for Overall Event E:**

$$P(E) = P(\text{Path } 1) + P(\text{Path } 2) + \cdots + P(\text{Path } k) = \sum_{i=1}^{k} P(\text{Path } i)$$

- *Verbal Description:* If a multi-step process can result in event E through k mutually exclusive sequential pathways, the total probability $P(E)$ equals the sum of the probabilities of each individual path.

- **Expanded Pathway Representation:**

$$P(E) = \sum_{i=1}^{k} P(A_i \cap E) = \sum_{i=1}^{k} P(A_i) \cdot P(E \mid A_i)$$

- *Verbal Description:* Each pathway probability is calculated using the Multiplication Rule ($P(A_i) \cdot P(E \mid A_i)$) across stages and summed across all disjoint pathways.

Day 8: Conditional Probability & Restricted Sample Spaces

1. Definition of Conditional Probability

- **Conditional Probability Formula:**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad \text{for } P(B) > 0$$

- *Verbal Description:* The conditional probability of event A given that event B has occurred equals the joint probability of both A and B divided by the probability of the conditioning event B.

- **Reduced Sample Space Formula (Equally Likely Outcomes):**

$$P(A \mid B) = \frac{N(A \cap B)}{N(B)}$$

- *Verbal Description:* In a sample space of equally likely outcomes, the conditional probability of A given B is the number of outcomes in the intersection of A and B divided by the total count of outcomes in the restricted sample space B.

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2. Rearranged Forms & Key Conditional Properties

- **Joint Probability via Conditional Probability:**

$$P(A \cap B) = P(B) \cdot P(A \mid B) = P(A) \cdot P(B \mid A)$$

- *Verbal Description:* The joint probability of A and B can be calculated by multiplying the probability of the given event by the conditional probability of the target event given the first.

- **Conditional Complement Rule:**

$$P(A^c \mid B) = 1 - P(A \mid B)$$

- *Verbal Description:* Given that event B has occurred, the conditional probability that event A does not occur equals 1 minus the conditional probability that event A occurs.

- **Conditioning Under Independence:**

$$P(A \mid B) = P(A) \quad \text{and} \quad P(B \mid A) = P(B) \quad (\text{for } P(A) > 0,\, P(B) > 0)$$

- *Verbal Description:* If events A and B are independent, knowing that event B occurred provides no information about event A, so the conditional probability equals the unconditional probability.

- **Conditional Set Difference Rule:**

$$P(A \cap B^c) = P(A) - P(A \cap B)$$

$$P(H \cap A^c) = P(H) - P(A \cap H)$$

- *Verbal Description:* The probability of a target event occurring without another event equals the total probability of the target event minus their joint intersection probability.

- **General Addition Rule Under Conditioning:**

$$P(A_1 \cup A_2 \mid B) = P(A_1 \mid B) + P(A_2 \mid B) - P(A_1 \cap A_2 \mid B)$$

- *Verbal Description:* All standard probability rules and axioms, including the general addition rule, hold within any restricted sample space B.

Day 9: Law of Total Probability & Partitioning Sample Spaces

1. Sample Space Partitioning

- **Partition Conditions:** A collection of events $\{B_1, B_2, \ldots, B_k\}$ forms a partition of sample space S if:

1. **Mutually Exclusive (Disjoint):** $B_i \cap B_j = \varnothing$ for all $i \neq j$ (i.e., $P(B_i \cap B_j) = 0$).
2. **Collectively Exhaustive:** $\bigcup_{i=1}^{k} B_i = B_1 \cup B_2 \cup \cdots \cup B_k = S$.

- *Verbal Description:* A partition divides the entire sample space into non-overlapping pieces that together cover all possible outcomes.

- **Sum of Partition Probabilities:**

$$\sum_{i=1}^{k} P(B_i) = P(B_1) + P(B_2) + \cdots + P(B_k) = 1$$

- *Verbal Description:* The probabilities of all mutually exclusive events in a partition must sum to exactly 1.

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2. The Law of Total Probability (LTP)

- **Joint Summation Form:**

$$P(A) = \sum_{i=1}^{k} P(A \cap B_i)$$

- *Verbal Description:* The unconditional probability of event A is obtained by summing the joint probabilities of A occurring with each partition piece B_i.

- **Standard Conditional Form (LTP Formula):**

$$P(A) = \sum_{i=1}^{k} P(B_i)P(A \mid B_i) = P(B_1)P(A \mid B_1) + P(B_2)P(A \mid B_2) + \cdots + P(B_k)P(A \mid B_k)$$

- *Verbal Description:* The total unconditional probability of event A is the weighted average of the conditional probabilities of A across each risk category or partition piece, weighted by the prior probability of each piece.

- **Two-Event Special Case (Single Event B and Complement B^c):**

$$P(A) = P(B)P(A \mid B) + P(B^c)P(A \mid B^c)$$

- *Verbal Description:* When partitioning by a single event B and its complement B^c, the total probability of A is $P(A \mid B)P(B)$ plus $P(A \mid B^c)P(B^c)$.

Day 10: Bayes' Theorem & Probability Tree Diagrams

1. Bayes' Theorem

- **General Partition Form:**

$$P(B_j \mid A) = \frac{P(B_j \cap A)}{P(A)} = \frac{P(B_j)P(A \mid B_j)}{\sum_{i=1}^{k} P(B_i)P(A \mid B_i)}$$

- *Verbal Description:* The posterior probability of partition slice B_j given that event A has been observed equals the joint probability of B_j and A divided by the total unconditional probability of event A (derived via LTP).

- **Two-Event Special Case:**

$$P(B \mid A) = \frac{P(B)P(A \mid B)}{P(B)P(A \mid B) + P(B^c)P(A \mid B^c)}$$

- *Verbal Description:* Updating the probability of hypothesis B after observing outcome A when the sample space is partitioned into B and B^c.

- **Posterior Probabilities Axiom:**

$$\sum_{i=1}^{k} P(B_i \mid A) = 1$$

- *Verbal Description:* The sum of updated (posterior) probabilities across all partition pieces conditioned on event A must equal 1.

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2. Probability Tree Diagram & Pathway Calculations

- **Key Terminology:**
  - **Prior Probability ($P(B_j)$):** Initial probability of event B_j before observing evidence.
  - **Likelihood ($P(A \mid B_j)$):** Conditional probability of observing evidence A given event B_j.
  - **Path Product / Joint Probability ($P(B_j \cap A)$):** Multiplying along a tree branch: $P(B_j \cap A) = P(B_j)P(A \mid B_j)$.
  - **Posterior Probability ($P(B_j \mid A)$):** Updated probability of B_j after observing evidence A.

- **Bayes' Calculation Formula for Tree Diagrams:**

$$P(B_j \mid A) = \frac{\text{Path Product of Interest}}{\text{Sum of All Path Products for Event } A}$$

  - *Verbal Description:* In a probability tree diagram, the posterior probability equals the joint path product of the specific scenario divided by the sum of path products for all branches leading to observed event A.